

Predict on New Image with a Trained Model

Purpose: Use the trained model to predict the class of a new image.

Code	Purpose	Explanation
<code>model.eval()</code>	Evaluation Mode	Sets the model to testing mode. Disables training-specific layers like Dropout and BatchNorm updates.
<code>input_image = preprocess_image("test.jpg")</code>	Load Test Image	Opens and preprocesses the image using the function created in Question 1.
<code>with torch.no_grad():</code>	Disable Gradient Calculation	Saves memory and speeds up prediction because no training is happening.

<code>output = model(input_image)</code>	Model Prediction	Passes the image through the trained model and gets prediction scores.
<code>output.argmax(1)</code>	Find Highest Score	Selects the class index with the highest prediction score.
<code>.item()</code>	Convert to Integer	Converts PyTorch tensor value into a normal Python integer.
<code>predicted_class = ...</code>	Store Prediction	Saves the predicted class label number.
<code>print("Predicted Class:", predicted_class)</code>	Display Result	Prints the predicted class.

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Step-by-Step Flow

Step	Operation	Example Output
1	Load image	<code>test.jpg</code>
2	Resize & preprocess	<code>Tensor [1, 3, 224, 224]</code>
3	Send to model	<code>model(input_image)</code>
4	Model outputs scores	<code>[0.10, 0.85, 0.05]</code>
5	Find maximum score	<code>0.85</code>
6	Get class index	<code>1</code>
7	Print result	<code>Predicted Class: 1</code>

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Understanding `argmax(1)`

Suppose the model outputs:

```
output = [[0.12, 0.75, 0.13]]
```

Class	Score
0	0.12
1	0.75
2	0.13

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```
predicted_class = output.argmax(1).item()
```

Result:

1

because **0.75** is the highest score.

What does `torch.no_grad()` do?

With <code>torch.no_grad()</code>	Without <code>torch.no_grad()</code>
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Faster
prediction

Slower

Less memory
usage

More memory
usage

Used for
testing

Used for training

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Overall Flow

test.jpg



preprocess_image()



[1,3,224,224]



model(input_image)



Prediction Scores

↓
argmax()
↓
Class Number
↓
Print Result

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Question

Answer

Why do we use
`model.eval()`?

To switch the model from training mode to prediction mode.

Why do we use
`torch.no_grad()`?

To disable gradient calculation during inference, making prediction faster and memory efficient.

What does
`argmax(1)` do?

Returns the index of the class with the highest prediction score.

What is
inference?

Using a trained model to predict the class of new unseen data.

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